CMPT 128 Introduction to Computing Science and Programming for Engineers	This course is a rigorous introduction to computing science and computer programming, suitable for engineering students. Students will learn fundamental concepts of computing science as well as develop basic skills in software development, with application to engineering problems. Topics - Computer Systems, Algorithms, Program design and development, Programming tools - Basic data types, Representation of values of variables, Conversion between data types - Variables, arithmetic, logical and relational operators, Input and Output, - Functions, arguments, return values, scope, - Control structures: decision and repetition structures - 1D and 2D Arrays, structures, strings - Dynamic memory allocation and pointers - Recursion, Searching and Sorting, Anallyzing and Comparing Algorithms (Big 0) Programming Language used: C and C++
ENSC 100W	This course is designed to provide an introduction to the practice of engineering, surveying its history and its current state. The social and political aspects of engineering decisions will be illustrated by a number of case studies.
ENSC 105W Process, Form, and Convention in Professional Genres	This course provides a general introduction to the principles of effective communication with special emphasis on the writing process, informative and persuasive writing, research papers, oral presentations, and resumes and cover letters. I wrote a persuasive paper on topic of laser weapons

ENSC 120	This introductory laboratory course will familiarize the students with operating electronics laboratory instrumentation such as linear power supply, digital multi-meter, function generator and oscilloscope.
MATH 150	**Chapter 1 - Functions and Models** 1.1 Four ways to represent a function 1.2 Mathematical Models: A Catalogue of Essential functions 1.3 New Functions from Old Functions 1.4 Exponential Functions 1.5 Inverse Functions and Logarithms **Chapter 2 - Limits and Derivatives** 2.1 Tangent and Velocity Problems 2.2 Limit of a Function 2.3 Calculating Limits Using the Limit Laws 2.5 Continuity 2.6 Limits at Infinity; Horizontal Asymptotes 2.7 Derivatives and Rates of Change 2.8 The Derivative as a Function **Chapter 3 - Differentiation Rules** 3.1 Derivatives of Polynomials and Exponential Functions 3.2 Product and Quotient Rules 3.3 Derivatives of Trigonometric Functions 3.4 The Chain Rule 3.5 Implicit Differentiation 3.6 Derivatives of Logarithmic Functions 3.7 Rates of Change in the Natural and Social Sciences 3.8 Exponential Growth and Decay 3.8 Newton's Law of Cooling 3.9 Related Rates 3.10 Linear Approximations and Differentials 3.11 Hyperbolic Functions (Optional)

	Chapter 4 - Applications of Differentiation 4.1 Maximum and Minimum Values 4.2 The Mean Value Theorem 4.3 How Derivatives Affect the Shape of a Graph 4.4 Indeterminate Forms and L'Hospital's Rule 4.5 Summary of Curve Sketching 4.7 Optimization Problems 4.8 Newton's Method **Chapter 10 - Parametric Equations and Polar Coordinates** 10.1 Curves Defined by Parametric Equations 10.2 Calculus with Parametric Curves 10.3 Polar Coordinates
ECON 103	This course is designed with two main goals in mind. The primary purpose of the course is to provide a sufficient understanding of the fundamental principles of economics. This includes propositions about individual behavior, output of firms, market structure, and the organization of economic activity. However, I also want to convince you that economics is an interesting field of study, and I intend to do this with a wide variety of examples. Throughout this course we will constantly apply economics to every aspect of your life. I hope to surprise you, perhaps offend you, and ultimately whet your intellectual appetite towards economics.
ENSC 180 Intro to Engineering Analysis	Introduction to MATLAB and its use in engineering. Implementation, verification, and analysis of various engineering algorithms used in signal and image processing, robotics, communications engineering.

MATH 152 Calculus II	**Chapter 5 - Integrals** 1. Areas and Distances 2. The Definite Integral 3. The Fundamental Theorem of Calculus 4. Indefinite Integrals 5. Substitution Rule **Chapter 6 -Applications of Integration** 1. Areas between Curves 2. Volumes 3. Volumes by Cylindrical Shells 5. Average Value of a Function (optional) **Chapter 7 -Techniques of Integration** 1. Integration by Parts 2. Trigonometric Integrals 3. Trigonometric Substitution 4. Integration of Rational Functions by Partial Fractions 5. Strategy for Integration 6. Integration Using Tables and Computer Algebra Systems 7. Approximate Integration 8. Improper Integrals **Chapter 8 - Further Applications of Integration** 1. Arc Length 2. Area of a Surface of Revolution **Chapter 10 - Parametric Equations and Polar Coordinates** 2. Calculus with Parametric Curves 4. Areas and Lengths in Polar Coordinates **Chapter 11 - Infinite Sequences and Series** 1. Sequences 2. Series 3. The Integral Test and Estimates of Sums 4. The Comparison Tests 5. Alternating Series

	6. Absolute Convergence and the Ratio and Root Tests 7. Strategy for Testing Series 8. Power Series 9. Representations of Functions as Power Series 10. Taylor and McLaurin Series 11. Applications of Taylor Polynomials **Chapter 9 - Differential Equations** 1. Modeling with Differential Equations 2. Direction Fields 3. Separable Equations 4. Models for Population Growth
MATH 232 Applied Linear Algebra	**Vectors** Vectors in Euclidean n-Space Dot Product and Orthogonality Lines and Planes **Systems of Linear Equations** Row Reduction (Gaussian elimination) to Echelon form The Geometry of Linear Systems Applications in business, science and engineering **Matrices** Matrix operations Matrix inverse; and properties of matrices Elementary matrices and calculating matrix inverses Matrices with special forms. **Linear Transformations** Matrices as transformations Geometry of Linear Transformations Kernel and range Composition and Invertibility Application to Computer Graphics (optional)

	Determinants Calculating determinants Properties of determinants Cramer's rule (optional) **Complex Numbers** Arithmetic in Cartesian co-ordinates. The complex plane, complex conjugate, magnitude and argument (phase). Polar form, De Moivre's formula and Euler's formula. Roots of quadratic polynomials. **Eigenvalues and Eigenvectors** Properties and geometry Complex eigenvalues and complex eigenvectors Dynamical Systems and Markov Chains Application to Economics: the Leontief model (optional) The Power Method; Application to Internet Search Engines Matrix Similarity and Diagonalization **Subspaces of R^n** Subspaces and Linear Independence Basis and Dimension The Fundamental Spaces of a Matrix Rank Change of basis **Orthogonality** Projection Orthogonal bases and the Gram Schmidt process Orthogonal matrices (optional) Application to least squares approximation
PHYS 120 Mechanics and Modern Physics	Motion in one, two and three dimensions. Newton's laws and applications, work, energy, momentum, collisions, circular motion, gravitation, and relativity. Rotational motion, torque, angular

	momentum. Special relativity.
CHEM 121 General Chemistry and Laboratory I	3 lecture hours/week; 1 tutorial hour/week; 4 lab hours on alternate weeks Lecture Topics: Atoms, Molecules, and Ions Stoichiometry Chemical Reactions Gases Thermochemistry Atomic Theory Bonding Concepts Covalent Bonding Orbitals Liquids and Solids Solutions Laboratory Experiments: Exp. 1 Introduction: Chemical Reactions and Equations I (first week of classes all students must attend) Exp. 2 Chemical Reactions and Equations II: Sequence of Copper Reactions Exp. 3 Redox Titrations: Analysis of Commerical Bleach Exp. 4 Molar Mass of a Gas Exp. 5 Thermochemistry: Use of Hess' Law Exp. 6 Spectrophotometric Analysis of Cobalt Ions in Solution Exp. 7 Colligative Properties: Freezing Point Depression
MATH 251 Calculus III	Rectangular, cylindrical and spherical coordinates. Vectors, lines, planes, cylinders, quadric surfaces. Vector functions, curves, motion in space. Differential and integral calculus of several variables. Vector fields, line integrals, fundamental theorem for line integrals, Green's theorem. Vectors and Geometry of Space: Three Dimensional Coordinate System Vectors The Dot Product The Cross Product

	Equations of Lines and Planes Cylinders and Quadric Surfaces Vector Functions: Vector Functions and Space Curves Derivatives and Integrals of Vector Functions Arc Length and Curvature Motion in Space Partial Derivatives: Functions of Several Variables Limits and Continuity Partial Derivatives Tangent Planes and Linear Approximations The Chain Rule Directional Derivatives and the Gradient Vector Maximum and Minimum Values Lagrange Multipliers and Constrained Maximum and Minimum Problems Multiple Integrals: Double Integrals over Rectangles Iterated Integrals Double Integrals over General Regions Double Integrals in Polar Coordinates Applications of Double Integrals Triple Integrals Triple Integrals in Cylindrical Coordinates Triple Integrals in Spherical Coordinates Change of Variables in Multiple Integrals Vector Calculus: Vector Fields Line Integrals The Fundamental Theorem for Line Integrals Green's Theorem
MATH 310 Intro to Ordinary Differential Equations	First Order differential equations Second- and higher-order linear equations Systems of first-order equations The Laplace transform Introduction to numerical methods and nonlinear systems Applications in the physical, biological and social sciences
PHYS 121 Optics, Electricity and Magnetism	Light waves, reflection, refraction, geometrical optics, interference, diffraction. Electric force, electric field, electrostatic

	potential. Capacitance, dielectrics, currents, DC circuits. Magnetic field, Biot-Savart Law, Ampere's Law, charge motion in electric and magnetic fields, torque on a current loop, Faraday's Law, electrical generators and motors, inductance, LCR circuits. Topics: 1. Electricity 2. Magnetism 3. Simple circuits 4. Optics 5. Topics from applied physics
ENSC 204 Graphical Communication for Engineering ENSC	An introduction to engineering graphical communication, Computer Aided Design (CAD) and workshop usage. Objectives are to improve students' literacy in the use of graphics to communicate engineering information, and their ability to visualize and create 3D models
ENSC 220 Electric Circuits I	Fundamental electrical circuit quantities, and circuit elements; circuits laws such as Ohm law, Kirchoff's voltage and current laws, along with series and parallel circuits; operational amplifiers; network theorems; nodal and mesh methods; analysis of natural and step response of first (RC and RL), as well as second order (RLC) circuits; real, reactive and rms power concepts. In addition, the course will discuss the worker safety implications of both electricity and common laboratory practices such as soldering.
ENSC 251 Software Design and Analysis for Engineers	Fundamentals for designing and implementing modular programs using a modern object-oriented programming language with a focus on understanding the performance implications of design choices on non-traditional computing platforms. Lecture topics include: classes; objects; debugging, testing & verification; design analysis & abstraction; error handling; fundamental data structures such as lists,

	trees, and graphs; and big-0 complexity analysis.computing platforms. Lecture topics include: classes; objects; debugging, testing & verification ; design analysis & abstraction ; error handling; fundamental data structures such as lists, trees, and graphs; and big-0 complexity analysis.
ENSC 252 Fundamentals of Digital Logic & Design	Design of digital systems. In particular, students will learn basic digital design concepts including the implementation of synthesizable combinational and sequential logic using HDL and computer based design tools to implement their designs on a FPGA. Used tools like Modelsim, Quartus and Altera Cyclone V development board with VHDL
CMPT 225 Data Structures and Programming	This course explores fundamental algorithms and data structures that can help in developing elegant and efficient solutions to complex problems. We will study their specification, analysis, implementation (in C++), experimental evaluation, and applications.
ENSC 225 Microelectronics I	Introduces the fundamentals of electronic devices with applications to active electronic circuits. Topics include physical structure and terminal characteristics of diodes and transistors; application of large and small signal device models in elementary amplifiers, current mirrors, and bias networks; behavioural models and frequency limitations of operational amplifiers.
ENSC 254 Introduction to Computer Organization	Covers the fundamentals of computer organization. Lecture topics include but not limited to: basics of computer systems, abstraction between hardware and software (using modern RISC-V instruction set), fundamentals of processor design and cache memory hierarchy design, and recent trends of parallel computing and domain-specific computing.
ENSC 280 Engineering Measurement and Data Analysis	Methods to collect and analyze engineering data. Topics include: engineering data representation, discrete and continuous probability density functions, engineering

	measurements, error analysis, test of hypotheses, linear and nonlinear regression, and design of experiments. This course includes a significant laboratory component comprising: laboratory measurements and statistical analysis of electronic circuits, introduction to electronic device behaviour, instrument noise.
ENSC 320 Electric Circuits II	Topics covered include: use of Laplace transform in circuit analysis, including poles and zeros, frequency response and impulse response: convolution as a method for computing circuit responses: resonant and bandpass circuits; magnetically coupled circuits; two port circuits; and filtering. Also includes a laboratory component dealing with the design and implementation of active filters.
CMPT 275 Software Engineering I	Introduction to software engineering techniques used in analysis/design and in software project management. The course centres on a team project involving requirements gathering, object analysis and simple data normalization, use-case-driven user documentation and design followed by implementation and testing. Additionally, there is an introduction to project planning, metrics, quality assurance, configuration management, and people issues.
ENSC 351 Embedded and Real Time System Software	Concentrates on the problems encountered when attempting to use computers in real time (RT) and embedded applications where the computer system must discern the state of the real world and react to it within stringent response time constraints. Both design methodology and practical implementation techniques for RT systems are presented. Although some hardware will be involved, it should be noted that this course concentrates on real time software.
ENSC 380 Linear Systems	The objectives of this course are to cover the modelling and analysis of continuous and discrete signals using linear techniques. Topics covered include: a review of Laplace transforms; methods for the basic modelling of physical systems; discrete and continuous

	convolution; impulse and step response; transfer functions and filtering; the continuous Fourier transform and its relationship to the Laplace transform; frequency response and Bode plots; sampling; the Z-transform.
ENSC 327 Communication Systems	Introduction to analog and digital communications systems. The main topics are: a review of Fourier Transform; the representation of bandpass signals; random signals in communications, including stationarity, ergodicity, correlation, power spectra and noise; amplitude and frequency modulation; circuits and techniques for modulation and demodulation; frequency division multiplexing; baseband digital communication; time division and multiplexing; an introduction to basic digital modulation schemes such as BPSK, FSK and QPSK. Laboratory work is included in this course.
ENSC 350 Digital Systems Design	Presents advanced topics in digital design such as advanced state machine concepts, asynchronous design, hardware description languages, bus interfacing and DSP architecture. It also covers both the architecture and programming or field programmable logic devices.
MACM 316 Numerical Analysis I	A presentation of the problems commonly arising in numerical analysis and scientific computing and the basic methods for their solutions.
CMPT 300 Operating Systems I	This course aims to give the student an understanding of what a modern operating system is, and the services it provides. It also discusses some basic issues in operating systems and provides solutions. Topics include multiprogramming, process management, memory management, and file systems.
CMPT 310 Introduction to Artificial Intelligence	Artificial intelligence is an area with vast application scenarios. This course is intended as a prerequisite for advanced-level AI courses and covers the fundamental

	concepts and foundational mathematical formulations in machine learning and AI. Topics include supervised learning, linear regression, gradient descent, search algorithms, Markov decision processes, and Bayes nets.
ENSC 429 Digital Signal Processing	Discrete time signals and systems, sampling and quantization. The Discrete Fourier Transform and fast transforms. Digital filters, IIR and FIR, design procedures and implementations. Quantization noise in digital filters and transforms. Random signals, the response to linear systems to random signals. Introduction to adaptive systems. Introduction to system architectures for digital signal processing. Laboratory work includes familiarization with digital signal processing software packages.
ENSC 410 The Business of Engineering	This course covers the business, management and entrepreneurial concepts that are important to engineers who manage projects, run businesses, or need to decide on the most efficient method for accomplishing a task. The topics to be covered include: financial accounting, rates of return, taxes, cost-benefit analyses, marketing, financing methods, and business plans.
ECON 105 Principles of Macroeconomics	The principal elements of theory concerning money and income, distribution, social accounts, public finance, international trade, comparative systems, and development and growth.
ENSC 406 Engineering Ethics, Law, and Professional Practice	Provides an introduction to the engineering profession, professional practice, engineering law and ethics, including the issues of worker and public safety. It also offers opportunities to explore the social implications and environmental impacts of technologies, including sustainability, and to consider engineers' responsibility to society. Special emphasis will be on technical communication skills. This course helps students successfully negotiate the transition to a professional career.

ENSC 324 Electronic Devices	The essential physics of silicon semiconductor devices that form the heart of integrated circuits today are covered. An introduction to semiconductor device physics upon which device models are based leading to the development of the drift-diffusion equations. The static and dynamic behavior of PN junction diodes, bipolar junction transistors, and field effect transistors will be covered along with the application of the developed device models to integrated circuit design.
MACM 201 Discrete Mathematics II	Topics covered include graph theory, trees, inclusion-exclusion, generating functions, recurrence relations, and optimization and matching.
ENSC 427 Communication Networks	This course covers the techniques needed to understand and analyze modern data communications networks using a top-down, layered approach to computer networking. It covers the basic architecture of packet networks and their network elements and the protocols used to enable transmission of packets. This is a project oriented undergraduate course. Students will be introduced to various tools for simulating communication networks such as ns-3. Laboratory work is included in this course.
ENSC 452 / 462 Advanced Digital System Design	Digital system design considerations including methodologies, specification, SoC partitioning, fault tolerance, design reuse, debugging and verification.
ENSC 405W Capstone A	Built the proof of concept and chassis for capstone B project
CMPT 354 Database Systems I	Logical representations of data records. Data models. Studies of some popular file and database systems. Document retrieval. Other related issues such as database administration, data dictionary and security. Used tools like mysql, xampp
ENSC 440 Capstone B	completed a capstone project called Robo Snacks, where my team and I developed an autonomous delivery robot designed to

	streamline indoor catering services. We built the system using a Jetson Nano and the ROS framework, implementing Gmapping SLAM to allow the robot to map and navigate complex floor plans in real-time without manual intervention. To make it user-friendly, we also developed a dedicated Android application that serves as the control interface for initiating deliveries. Our final prototype successfully demonstrated fully autonomous movement and obstacle avoidance, addressing a multi-billion dollar market in the food service industry.
CMPT 454 Database Systems II	An advanced course on database systems which covers crash recovery, concurrency control, transaction processing, distributed database systems as the core material and a set of selected topics based on the new developments and research interests, such as object-oriented data models and systems, extended relational systems, deductive database systems, and security and integrity.
ENSC 383 Feedback Control Systems	This course is an introduction to the analysis, design, and applications of continuous time linear control systems. Topics include transfer function representation of open and closed loop systems, time domain specifications and steady state error, sensitivity analysis, time and frequency response, and stability criteria. It includes a treatment of methods for the analysis of control systems based on the root locus, Bode plots and Nyquist criterion, and their use in the design of PID, and lead-lag compensation. Lab work is included in this course.
ENSC 450 VLSI Systems Design	An introduction to the design of Very Large Scale Integrated (VLSI) circuits and systems (System-on-Chip, SoC) using mainly CMOS technology. SoC design techniques and applications will be covered. Basic topics will include: CMOS technology and circuit layout rules; combinational and sequential logic; logic simulation; systems design; design for verification and testability; and embedded-processor design and application. An advanced digital design flow based on the VHDL hardware description language will be

	introduced and exercised in the labs.